Factors Affecting Commuter Choice

Course Project - 6

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Introduction to Multi-Modal Mobility Systems

in

M.Tech Programme

by

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Abstract

This study investigates the key factors influencing commuter choice among various transport modes in Bengaluru, including buses, metro, private two-wheelers, cars, cab/ride-hailing services, walking, and cycling. A structured stated and revealed preference survey was designed with more than eighteen questions to capture travel behaviour, service quality perceptions, and challenges associated with different modes. Data were collected from at least 30 participants, with a minimum of 20 respondents unaffiliated with IISc, as required by the project guidelines. The collected data were analyzed using descriptive statistics, cross-tabulations, and visualizations to identify patterns in commuter behaviour, barriers to multimodal integration, and determinants that influence the adoption of sustainable transport modes. The results highlight the importance of factors such as travel time reliability, first/last-mile connectivity, comfort, cost, and safety in shaping mode choice decisions. Based on these insights, the study provides evidence-based recommendations to improve multimodal accessibility, enhance the attractiveness of public transport, and encourage a shift toward sustainable mobility in Bengaluru. Limitations of the study and directions for future research are also discussed.

Chapter 1

Introduction

Urban transportation systems are fundamental to ensuring mobility, accessibility, and economic productivity within rapidly expanding urban centers. Bengaluru, marked by high population density and extensive daily travel requirements, is experiencing increasing challenges including severe road congestion, rising private vehicle ownership, inadequate multimodal coordination, and limited public transport penetration. Understanding the factors that shape commuter mode selection is vital for formulating policies that encourage sustainable transportation alternatives and reduce dependence on personal vehicles.

Commuter mode choice is influenced by a combination of socioeconomic characteristics, trip attributes, and perceptions related to convenience, affordability, accessibility, safety, and service reliability. Although public transport modes such as BMTC buses and Namma Metro provide essential connectivity, many commuters continue to prefer private vehicles due to first- and last-mile connectivity gaps, overcrowding, long waiting durations, limited integration between services, and the absence of unified ticketing systems.

This research examines the key determinants affecting commuter decisions among available travel modes in Bengaluru. A structured questionnaire survey was conducted at bus stops, metro stations, and through personal networks to capture diverse travel patterns from both public transport users and non-users. The survey evaluates demographic characteristics, trip behavior, reasons for adopting or avoiding public transport, challenges encountered during multimodal travel, and satisfaction levels related to service quality and system integration.

Insights derived from the collected data are expected to support evidence-based recommendations aimed at improving multimodal connectivity, enhancing the reliability of public transport, and promoting sustainable commuter behavior in Bengaluru.

Chapter 2

Methodology

This research employed a primary data collection strategy using a structured questionnaire survey to investigate the determinants influencing commuter mode choice in Bengaluru. The survey instrument consisted of multiple-choice and Likert-scale questions designed to gather information on demographic attributes, trip characteristics, public transport usage patterns, multimodal connectivity challenges, and satisfaction levels with existing services.

2.1 Survey Administration

Data collection was carried out through both field-based and online distribution methods to ensure diversity and representativeness among respondents. Surveys were conducted across the following locations and groups:

- **Bus Stops and BMTC Terminals:** Surveys were administered at major high-demand bus stops and interchange points during peak periods (8:00–11:00 AM and 5:00–8:00 PM) to capture real-time commuter experiences related to waiting times, crowding levels, boarding processes, and service reliability.
- **Metro Stations (Namma Metro):** Respondents were approached at key metro stations and entry/exit hubs to examine multimodal transfer behavior, first- and last-mile connectivity constraints, parking availability, and integrated ticketing considerations.
- **Residential Areas and Personal Networks:** Additional responses were collected from students, working professionals, and acquaintances across residential neighborhoods. This enabled the inclusion of private vehicle users and individuals who do not rely on public transport, providing a balanced representation.

2.2 Sampling Strategy

A convenience sampling method was adopted due to time and accessibility constraints. Efforts were made to include varied demographic and socioeconomic populations. A minimum target of 30 participants was established in accordance with project requirements, and more than 20 responses were collected from individuals unaffiliated with IISc, ensuring representation beyond academic commuters.

2.3 Survey Structure and Importance of Question Groups

The questionnaire was structured into three principal sections, each addressing different analytical objectives:

1. **Demographic and Household Profile:** Included variables such as age, income, vehicle ownership, and family size, enabling assessment of socioeconomic influence on mode choice—particularly regarding affordability, accessibility, and comfort preferences.

2. **Trip Characteristics:** Questions regarding travel purpose, frequency, duration, distance, and cost provided insight into mobility patterns and pain points that affect commuter decision-making.

3. **Public Transport Usage, Accessibility, and Challenges:** Items assessing satisfaction levels, multimodal connectivity constraints, first- and last-mile linkages, and reasons for avoiding public transport allowed identification of critical barriers such as:

    - Inadequate last-mile connectivity
    - Excessive waiting times
    - Poor scheduling and route coverage
    - Comparative convenience of private transport

Chapter 3

Results and Discussion

Collected responses were compiled into a CSV dataset and analyzed using descriptive statistical methods, graphical visualization, and cross-tabulation to explore correlations between commuter characteristics and mode selection patterns. Insights derived from the analysis form the basis for actionable recommendations aimed at enhancing multimodal integration, improving accessibility to public transport, and promoting sustainable commuter behavior within Bengaluru.

3.1 Descriptive Statistics

3.1.1 Demographics

The survey sample reflects a predominantly young, educated, urban commuter population. Individuals aged 18–35 years constitute 90% of respondents, with 47% belonging to the 18–25 age group and 43% to the 26–35 group. The gender distribution comprises 67% male, 30% female, and 3% non-binary participants. Educational attainment is high, with 83% holding undergraduate or postgraduate degrees.

Household income levels indicate that 37% of respondents earn between 30,001–60,000 per month, followed by 30% earning between 60,001–1,00,000. Household size is moderate in most cases, with 57% reporting 3–4 members. Vehicle ownership is substantial, with 60% owning at least one private vehicle. These demographic characteristics are illustrated in Figure 3.1.

3.1.2 Travel Behavior

Trip purposes are evenly distributed between education (33%) and work (33%), with 23% related to social or leisure activities. Namma Metro is the most preferred travel mode (43%), followed by BMTC buses (17%), two-wheelers (13%), and cab/ride-hailing services (13%).

A majority (53%) travel daily, indicating routine commuting. Travel time is considerable, with 53% spending more than 46 minutes on one-way trips, and 23% exceeding 60 minutes. Trip distances mostly fall within 8–25 km (60%). These travel behaviour patterns are summarized in Figure 3.2.

3.1.3 Trip Purpose Patterns

Education-related travel heavily relies on public transport: 50% use Metro and 30% use BMTC buses. Work-related travel shows greater variation, with 30% using personal two-wheelers and 30% using auto-rickshaws. Social and leisure travel prioritizes flexibility and comfort, with cab services (43%) and Metro (57%) dominating. These patterns are shown in Figure 3.3.

3.1.4 Public Transport Satisfaction

The mean satisfaction rating for public transport is 3.07 out of 5, indicating overall neutral perception. Of all respondents, 53% rated 3, 27% rated 4–5 (satisfied), and 20% rated 1–2

(dissatisfied). Satisfaction by mode is visualized in Figure 3.3.

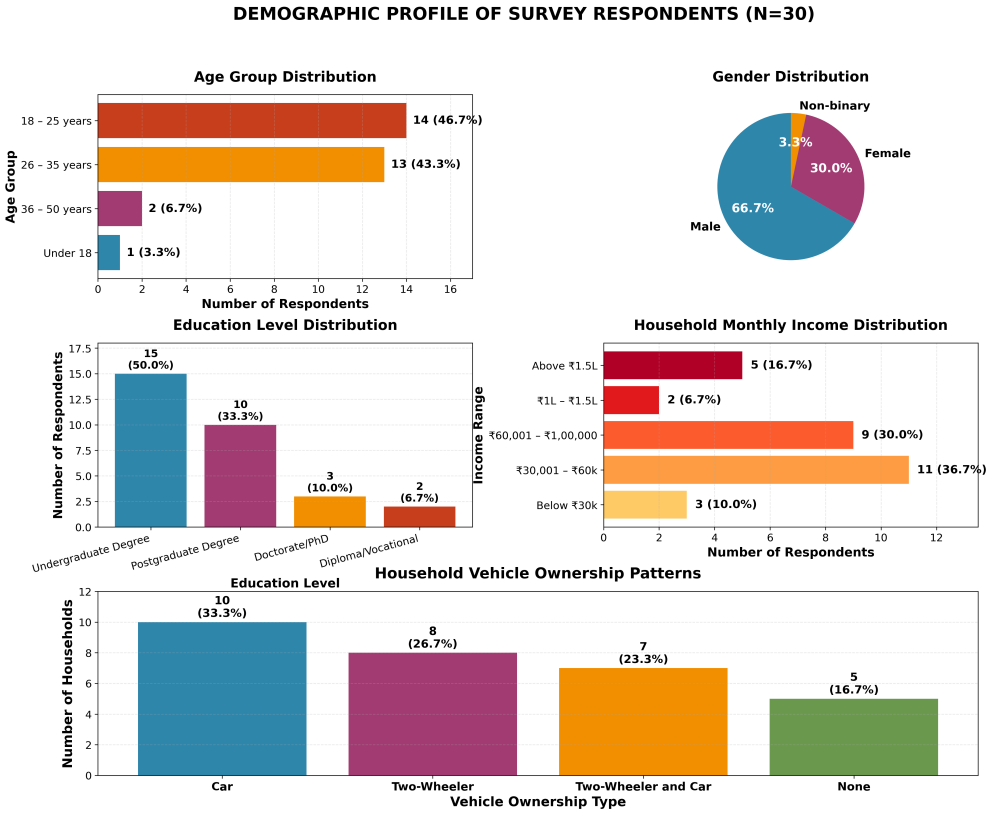


Figure 3.1: Demographic profile of survey respondents (N=30).

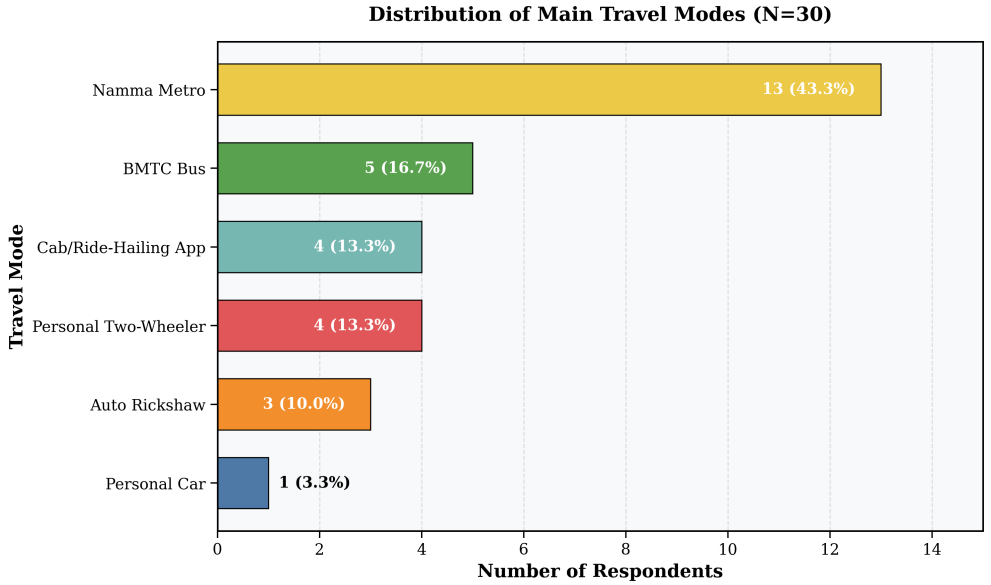


Figure 3.2: Travel mode distribution and travel behaviour characteristics (N=30).

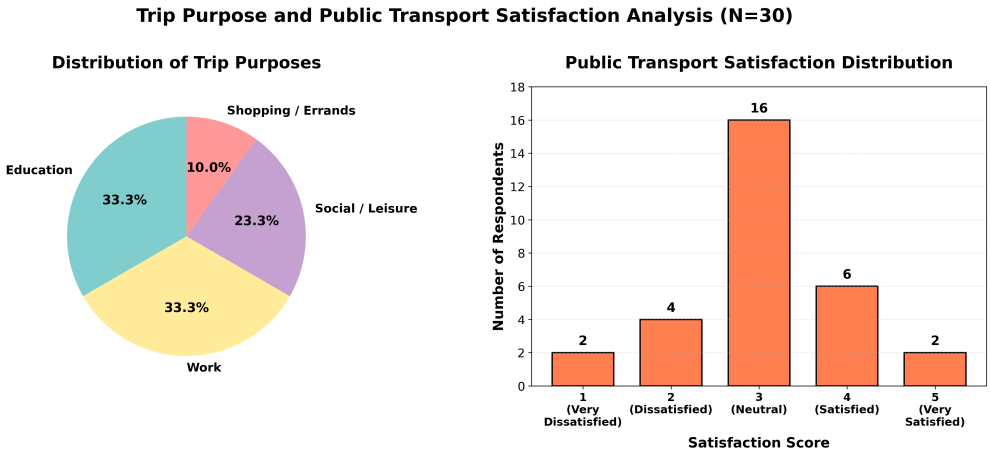


Figure 3.3: Trip purpose patterns and satisfaction levels across transport modes (N=30).

3.1.5 Cross-Tabulation Insights

Travel Mode by Demographics

Income strongly influences mode choice. Higher-income groups (above INR 1.5 L per month) report an 80% preference for Namma Metro, whereas middle-income respondents (INR 30k–INR 60k) show a more balanced distribution across Metro (45%), two-wheelers (27%), and other transport modes. Age also correlates with preferences: individuals aged 18–25 rely more on Metro (43%) and BMTC buses (29%), while the 26–35 group demonstrates more varied usage, including ride-hailing services (23%). These patterns are illustrated in Figure 3.4.

Satisfaction by Travel Mode

BMTC users report the highest satisfaction (3.80/5), despite lower usage share. This suggests that service frequency and availability, rather than quality, limit its adoption. Metro, cab services, and private cars average moderate satisfaction (3.0/5). Meanwhile, personal two-wheelers (2.75/5) and auto rickshaws (2.67/5) reflect lower satisfaction. These results are shown in Figure 3.5.

Vehicle Ownership Impact

Notably, 83% of vehicle owners still use public transport or shared mobility options. Among Metro users, 46% own cars and 23% own both car and two-wheeler, confirming that vehicle ownership does not eliminate demand for mass transit. This relationship is shown in Figure 3.6.

3.1.6 Summary of Travel Pattern Analysis

The integrated analysis of travel time, distance, cost, and mode choice highlights systemic mobility trends among Bengaluru commuters. Metro is the most widely used mode (43%), including for longer journeys, while BMTC primarily supports mid-distance travel. More than half of respondents (53%) spend over 46 minutes per one-way trip, consistent with typical Bangalore congestion patterns.

Trip distance results show that 60% of travel occurs within 8–25 km, matching cross-city commuting trends. Despite a 60% private vehicle ownership rate, strong Metro usage shows

commuters prefer public transit when it's reliable. Cost analysis reveals that 40% spend over 200 per trip—significant for middle-income groups.

The scatter plot of travel time vs. distance (Figure 3.7) shows clustering around 11–20 km and 30–60 minutes, typical of Bangalore commutes. Metro users display predictable travel times due to grade-separated operations, while cab and private vehicle trips show higher variability due to congestion. Instances of unexpectedly long Metro trips highlight poor first/last-mile connectivity, walking conditions, and transfer delays.

These insights emphasize the need for multimodal integration, better local accessibility, and unified ticketing to shift travel behavior toward sustainable modes.

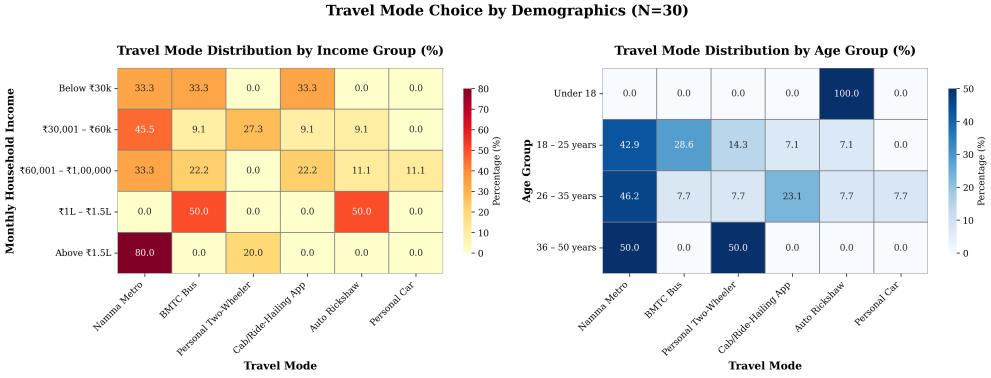


Figure 3.4: Travel mode choice segmented by demographic factors (N=30).

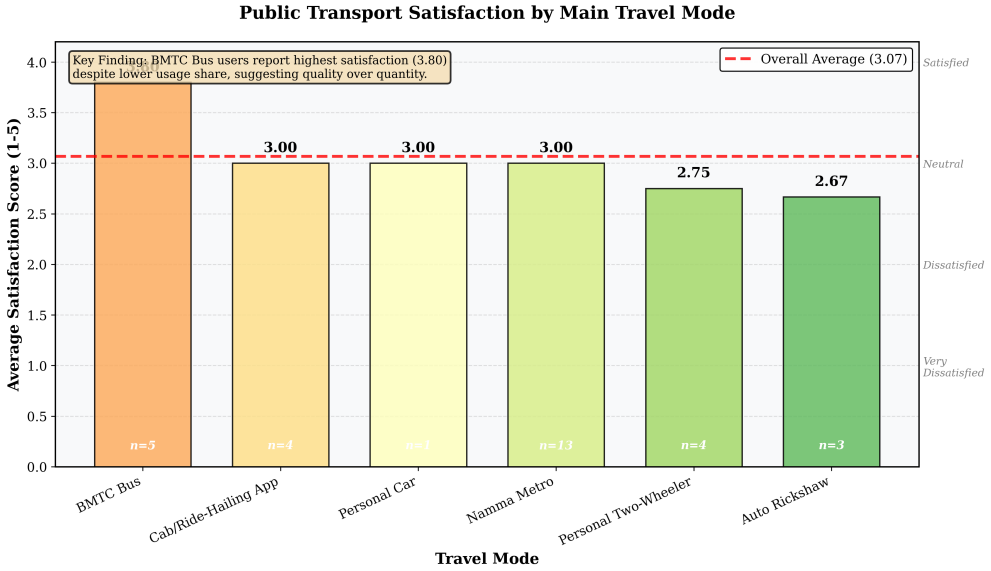


Figure 3.5: Public transport satisfaction levels across travel modes (N=30).

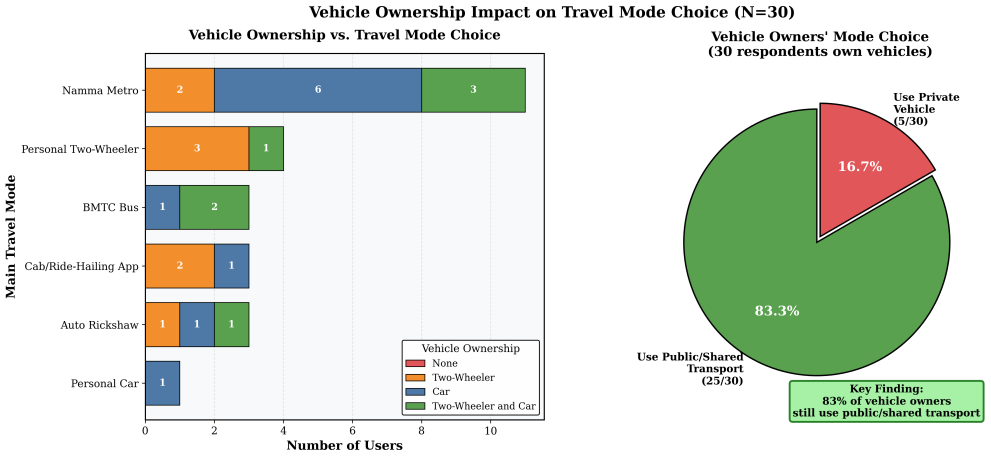


Figure 3.6: Impact of vehicle ownership on preferred travel modes (N=30).

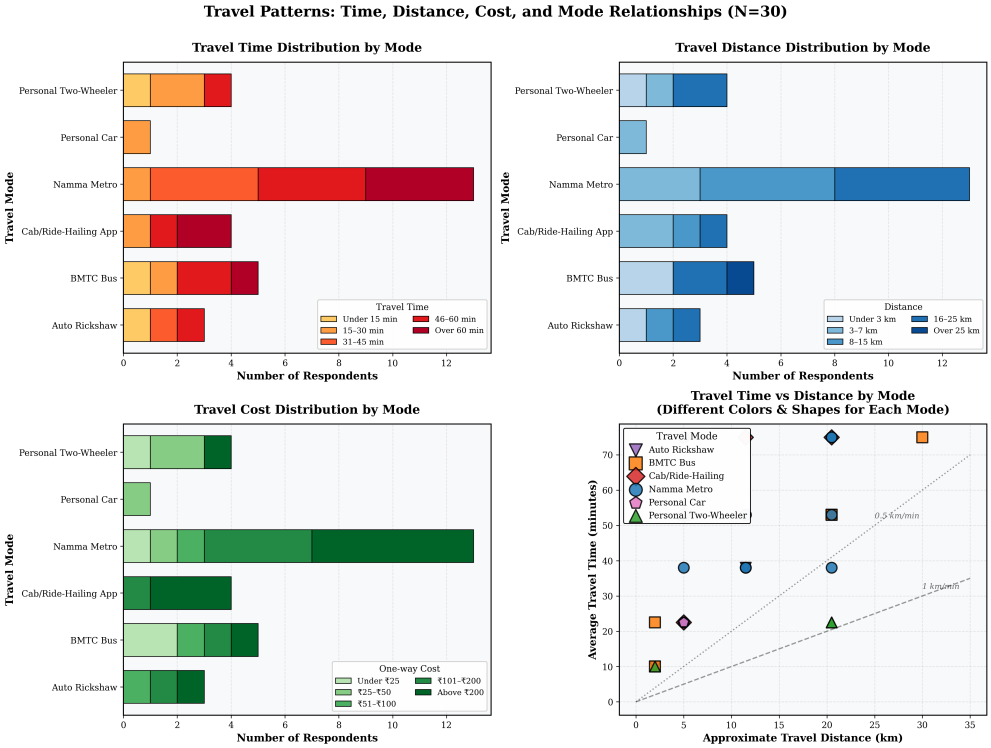


Figure 3.7: Travel pattern analysis: time, distance, cost, and mode relationships (N=30).

3.2 Recommendations

Based on the survey findings, the following focused recommendations are proposed:

- **Improve First/Last-Mile Connectivity:** Introduce reliable feeder services and enhance walking/cycling access, addressing the frequently reported lack of connectivity and poor pedestrian conditions.
- **Increase BMTC Bus Frequency:** The most common suggestion from respondents, aimed at reducing long waiting times and improving reliability.
- **Implement Common Ticketing:** Integrate BMTC, Metro, and shared mobility payments to enable seamless multimodal transfers.
- **Upgrade Pedestrian Infrastructure:** Remove footpath encroachments, improve lighting, and create continuous sidewalks to address major accessibility challenges.
- **Regulate Auto-Rickshaw and Ride-Hailing Zones:** Provide designated pickup/drop areas to reduce curbside conflicts and improve user experience.
- **Develop Protected Cycling Lanes:** Support safe first/last-mile movement and encourage low-emission travel options.

3.3 Limitations

This study has several limitations:

- **Small Sample Size (N=30):** Results are indicative rather than generalizable.
- **Convenience Sampling Bias:** Over-representation of young and educated respondents limits demographic diversity.
- **Geographic and Mode Bias:** Limited representation of suburban, cycling, and walking-only commuters reduces modal breadth.
- **Self-Reported Data:** Travel time, distance, and satisfaction values may include perception and recall biases.
- **Single-Day Data Collection:** Does not capture variations across different days or peak periods.

Appendix: Supplementary Information

This appendix provides a brief summary of the survey instrument; the full questionnaire is submitted as a separate document as required in the project guidelines.

A. Survey Instrument (Summary)

The survey instrument used in this project was designed to collect stated and revealed preference data related to commuter behaviour in Bengaluru. The survey contained more than fifteen structured questions covering the following themes:

- **Demographics:** age group, gender, education, income, household size, vehicle ownership.
- **Trip Characteristics:** purpose, frequency, travel time, distance, cost, primary mode.
- **Public Transport Usage:** frequency of use, reasons for non-use, satisfaction ratings including service frequency, reliability, comfort, safety, cleanliness, and modal integration.
- **First/Last-Mile Connectivity:** access modes (walking, cycling, auto, cab, feeder), and key barriers such as poor footpaths, safety concerns, and inconvenient transfers.

Most responses were recorded using multiple-choice formats and 5-point Likert scales. One open-ended question captured improvement suggestions.

B. Supporting Material

The following elements used in the analysis are provided separately in the submission ZIP:

- Survey dataset (.csv)
- Full survey instrument (.pdf)

Summary statistics, cross-tabulations, and all figures presented in the main report were generated directly from the submitted dataset.